

The Regional Blood Bank Distribution Problem

Background

A national health authority manages **4 regional blood processing centers** and must supply **6 hospitals** across the region. Each day, processed blood units must be allocated from centers to hospitals before a strict morning deadline. Due to perishability, **unused stock is discarded** — making over-supply costly and under-supply dangerous.

The Setup

Supply (Blood Processing Centers)

Center	Location	Daily Capacity (units)
C1	Pretoria	120
C2	Johannesburg	95
C3	Durban	110
C4	Cape Town	75

Total Supply: 400 units/day

Demand (Hospitals)

Hospital	Location	Daily Requirement (units)
H1	Polokwane	60
H2	Nelspruit	45
H3	Bloemfontein	70
H4	East London	55
H5	Port Elizabeth	80
H6	Kimberley	50

Total Demand: 360 units/day

Transportation Costs (Rands per unit)

	H1	H2	H3	H4	H5	H6
C1	12	18	22	31	38	25
C2	15	14	20	28	35	22
C3	34	29	24	16	14	27
C4	41	37	26	22	18	30

Part A — Formulation (30 marks)

1. Define all **decision variables** clearly.
 2. State the **objective function** mathematically.
 3. Write out all **supply constraints**, **demand constraints**, and **non-negativity conditions**.
 4. Identify whether this problem is **balanced or unbalanced**, and explain what adjustment is needed before applying the transportation method.
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Part B — Initial Basic Feasible Solution (35 marks)

5. Apply the **Northwest Corner Method** to find an initial solution. Show the full allocation table at each step.
 6. Apply **Vogel's Approximation Method (VAM)** to find a second initial solution. Show all penalty calculations at every iteration.
 7. Compare the **total costs** of both initial solutions. Which gives a better starting point, and why is this relevant for the simplex iterations that follow?
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Part C — Optimality & Sensitivity (35 marks)

8. Apply the **MODI Method (UV Method)** to the VAM solution to test for optimality. Show all u and v values, all reduced costs, and identify any entering variable if the solution is not yet optimal. Perform one iteration if needed.
 9. Suppose C3 reports a **10% reduction in daily capacity** due to equipment failure. Reformulate the supply constraints and discuss how this changes the balance condition.
 10. The cost of shipping from C1 to H1 increases from **R12 to R20** due to a fuel levy. Without re-solving from scratch, use your knowledge of the optimal basis to reason about whether the current optimal solution is still valid. What concept from sensitivity analysis applies here?
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Bonus Question (10 marks)

A statistician claims that daily hospital demand follows a **Poisson distribution** with rates equal to the values in the demand table. Critically evaluate what this assumption implies for the deterministic model you built. Under what conditions would a **stochastic programming** or **simulation approach** be more appropriate than the LP formulation used here?

All answers must show full working. Unsupported answers will receive no marks.